

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import warnings
warnings.filterwarnings("ignore")
%matplotlib inline
```

```
In [2]: df = pd.read_csv("train.csv")
```

```
In [3]: df.shape
```

```
Out[3]: (891, 12)
```

```
In [4]: df
```

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/ 2117
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 1759
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O. 310128
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	11380
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	37345
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	21153
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	11205
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./ 660
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	11136
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	37037

891 rows × 12 columns

In [5]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

```
In [6]: df.describe()
```

```
Out[6]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	3
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	4
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	0
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	0
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	3
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	5

```
In [7]: df.isna().sum()
```

```
Out[7]: PassengerId    0
Survived             0
Pclass               0
Name                 0
Sex                  0
Age                 177
SibSp                0
Parch                0
Ticket              0
Fare                 0
Cabin              687
Embarked             2
dtype: int64
```

```
In [8]: df["Age"] = df["Age"].fillna(df["Age"].mean())
```

```
In [9]: df.isna().sum()
```

```
Out[9]: PassengerId      0
        Survived        0
        Pclass          0
        Name            0
        Sex             0
        Age             0
        SibSp           0
        Parch           0
        Ticket          0
        Fare            0
        Cabin          687
        Embarked        2
        dtype: int64
```

```
In [10]: def fun1(value):
        if (value == "male"):
            return 1
        else:
            return 0
```

```
In [11]: def fun1(value):
        if (value == "male"):
            return 1
        else:
            return 0
```

```
In [12]: def fun2(value):
        if (value == 'S'):
            return 0
        elif (value == 'C'):
            return 1
        elif (value == 'Q'):
            return 2
        else:
            return 0
```

```
In [14]: df["Sex"] = df["Sex"].apply(fun1)
```

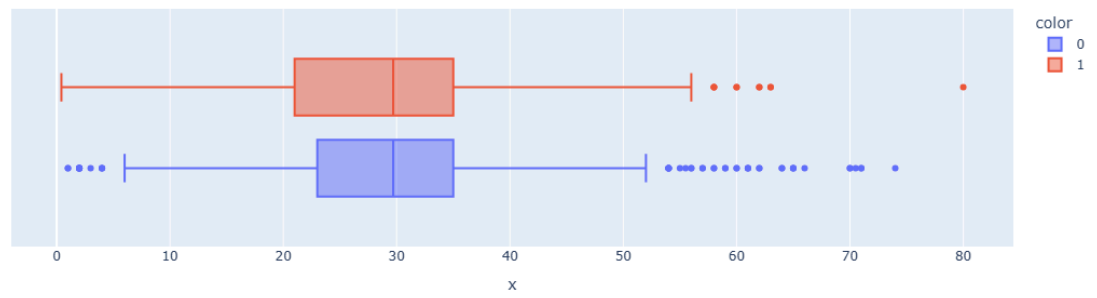
```
In [15]: df["Embarked"] = df["Embarked"].apply(fun2)
```

```
In [16]: df = df.drop("Cabin", axis=1)
```

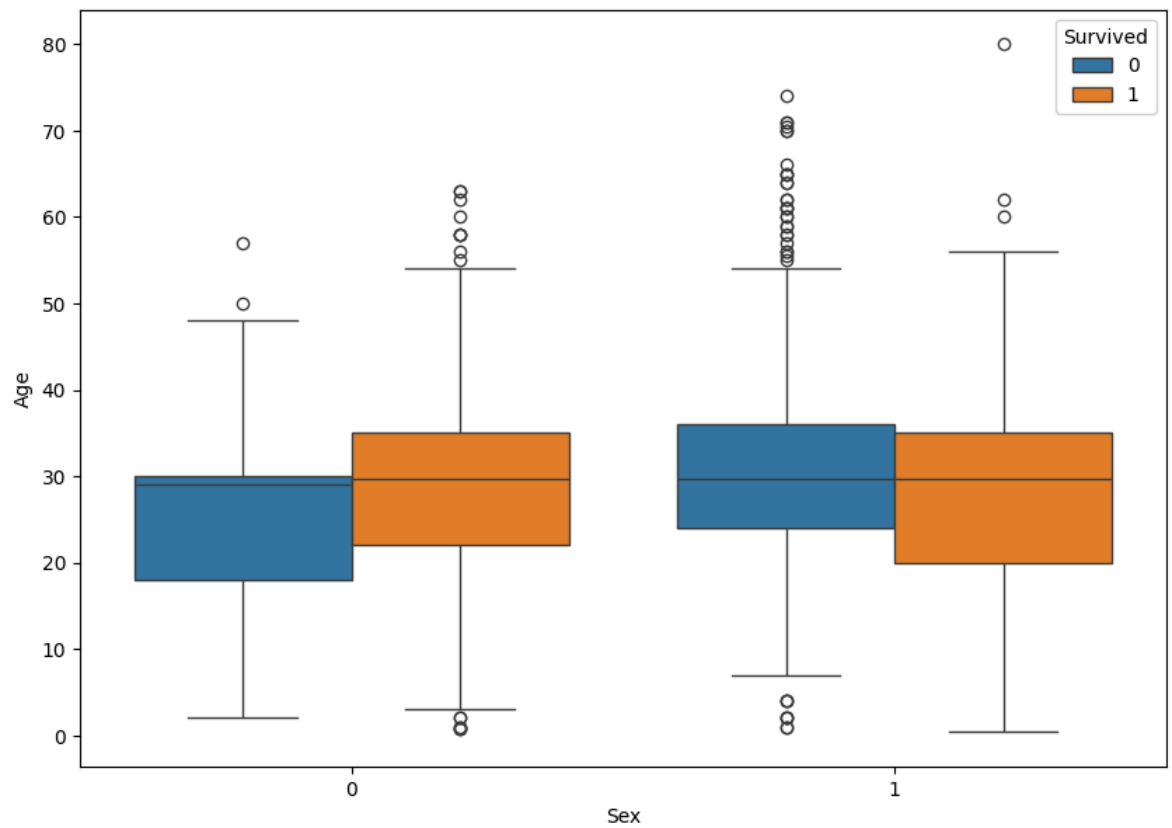
```
In [17]: df.shape
```

```
Out[17]: (891, 11)
```

```
In [18]: px.box(df["Sex"], df["Age"], color=df["Survived"])
```



```
In [19]: plt.figure(figsize=(10,7))
box = sns.boxplot(df,x="Sex", y="Age", hue="Survived")
plt.show()
```



```
In [ ]:
```